

Qualifying Exam in Microeconomics
Arizona State University
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There are four questions and you have four hours to complete the exam.

1. Consider a consumer with a utility function $U(x)$ defined on her consumption set, the positive orthant $\mathbb{R}_+^L = \{x \in \mathbb{R}^L | x \geq 0\}$. Suppose that $U(x)$ is strictly increasing, continuous, and strictly (quasi)concave. Prove that for any strictly positive x^* , there is an initial endowment ω and a non-negative, non-zero price vector, p , such that x^* is utility maximizing on the budget constraint determined by ω and p .

2. The consumption set X for goods traded in competitive markets is a nonempty, closed, and convex subset of $\mathbb{R}_+^L$ for $L \geq 2$ goods. (In particular the set X is not necessarily a lattice when endowed with the partial order $\geq$, the usual vector inequality on $\mathbb{R}^L$.) The consumer has preferences over $X \times Q$ represented by a function $u : X \times Q \rightarrow \mathbb{R}$ where Q is a nonempty, real interval. Points $q \in Q$ can be thought of as quantities of a good not traded on a market, for example a public good. Given $q \in Q$ and a strictly positive price-wealth pair (p, w) , the consumer chooses $x \in B(p, w) = \{x \in X | p \cdot x \leq w\}$ to maximize $u(x, q)$. The preference representation is continuous and locally nonsatiated on $X \times Q$. In what follows fix a price-wealth pair $(p, w) \gg 0$.

(a) Let $V(p, w, q)$ be the consumer's indirect utility function. Prove that if u is concave on $X \times Q$, then $V(p, w, \cdot)$ is concave on Q .

(b) Specialize now to the $L = 2$ case and impose the following strict-single crossing property:

If $(x'_1, -x'_2) > (x_1, -x_2)$, $q' > q$, and $u(x', q) \geq u(x, q)$, then $u(x', q') > u(x, q)$. (SSCP)

Let $q_1 > q_0$, let x^t solve the consumer's problem at $q = q_t$ for $t = 0, 1$. Does it follow that $x_1^1 \geq x_1^0$? If so, prove it; if not, identify additional conditions and explain why your set of conditions is sufficient for the conclusion. (Suggestion: set $U(x_1, q) = u(x_1, (w - p_1 x_1)/p_2, q)$.)

3. Consider the following pure exchange economy. There are m goods and a finite set A of agents. The positive orthant $\mathbb{R}_+^m$ is the consumption set of all agents; each agent $a \in A$ has endowment $\omega(a)$ in $\mathbb{R}_+^m$. Each agent a has preferences $\succ_a$ that are continuous, irreflexive ($x \not\succ_a x$) and satisfy

(a) $x \gg y \implies x \succ y$

(b) $x \gg y$ and $y \succ z$ implies $x \succ z$

Let $D(p, a) = \{x \in \mathbb{R}_+^m : p \cdot x \leq p \cdot \omega(a) \text{ and } [y \succ_a x \implies p \cdot y > p \cdot \omega(a)]\}$, that is to say, $D(p, a)$ is the demand set of agent a at prices p .

Let $\Psi(x, p, a) = \|p \cdot (x - \omega(a))\| + \|\inf\{p \cdot (y - \omega(a)) : y \succ_a x\}\|$.

Prove that for any $p \gg 0$, $\Psi(x, p, a) = 0 \implies x \in D(p, a)$

4. Consider an industry with two firms selling identical products. Each firm has a constant-returns technology with unit cost of $k > 0$ and the market inverse demand is $\max\{a - q, 0\}$, where $a > 2k > 0$ and q is aggregate output. (The inequality $a > 2k$ is to ensure that subgame-perfect output choices are positive.)

The firms are owned by different individuals—firm i by owner i —and, to produce any output at all, an owner must hire precisely one manager to decide how much output to produce. The (total) wage that owner i pays manager i is restricted to be of the form $w_i = \alpha_i + P_i(\varepsilon)$, where $\alpha_i \in \mathbb{R}$ is a fixed payment and $P_i(\varepsilon)$ is the “performance” of the manager defined to be a weighted average of profit π_i and revenue R_i ; specifically

$$P_i(\varepsilon) = \varepsilon_i R_i + (1 - \varepsilon_i) \pi_i \quad \leftarrow \varepsilon_i R_i + (1 - \varepsilon_i) (R_i - \text{cost})$$

where $\varepsilon_i \in [0, 1]$. (Profit π_i and revenue R_i are of course functions of the output pair (q_1, q_2) .) Each manager just cares about the wage paid, the higher the better, and each has an outside option that pays $\bar{w} > 0$.

The timing is that the owners first independently and simultaneously choose the contract parameters $(\alpha_i, \varepsilon_i) \in \mathbb{R} \times [0, 1]$ for $i = 1, 2$. Each manager observes both contracts. The managers then simultaneously and independently decide whether to accept or reject the contracts. After observing the acceptance decisions, any manager that has accepted a contract chooses an output (simultaneously with the other manager if both accepted contracts). If a manager rejects, then that firm’s output is zero and that manager collects the outside wage of $\bar{w}$. After any outputs are chosen, the managers who accepted contracts are paid according to the contract terms and their performance.

- (a) Identify the strategy set for manager 1 in this extensive form game. *horrible notation*

In what follows confine your attention to subgame perfect Nash equilibria in which each manager i accepts a contract $(\alpha_i, \varepsilon_i)$ whenever indifferent between accepting and rejecting it.

- (b) Consider any output-choice subgame after both contracts have been accepted. Calculate the output choices implied by subgame perfection. How does a change in α_i affect Nash equilibrium outputs in these subgames, assuming both managers still accept the contracts? Explain.

- (c) What must w_i equal in a subgame perfect Nash equilibrium in which both managers accept contracts? Explain. *why*

- (d) Fix $\varepsilon_2 = 0$, $\varepsilon_1 \in [0, 1]$ and $(\alpha_1, \alpha_2) \in \mathbb{R}^2$. Suppose both contracts are accepted. Confirm that firm 1’s profit in a Nash equilibrium of the output-choice subgame equals

$$\pi_1^*(\varepsilon_1) = \left(\frac{a - k + 2k\varepsilon_1}{3} \right) \left(\frac{a - k - k\varepsilon_1}{3} \right).$$

(The first term is firm 1’s Nash equilibrium output in the subgame and the second is the equilibrium difference between price and unit cost.) If you have trouble verifying this fact, then simply assume it in what follows.

- (e) Consider any subgame perfect Nash equilibrium in which both offered contracts are accepted. Prove that $\varepsilon_1 = \varepsilon_2 = 0$ is not part of such a SPNE. What is the economic significance of this fact?